



## ValueError: Input 0 of layer "model\_12" is incompatible with the layer: expected shape=(None, 256, 256, 3), found shape=(256, 256, 3)

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I am following [this](#) keras example with my own dataset, which has 3 classes. I load the images using

```
train = tf.keras.utils.image_dataset_from_directory(cloud_dir, labels = "inferred",
label_mode="categorical", batch_size=None)
```

I am not using a validation set nor a batch size. I am also augmenting and prefetching the data set by

```
data_augmentation = keras.Sequential(
    [
        keras.layers.RandomFlip("horizontal"),
        keras.layers.RandomRotation(0.1),
    ]
)

train = train.map(
    lambda img, label: (data_augmentation(img), label),
    num_parallel_calls=tf.data.AUTOTUNE,
)

train = train.prefetch(tf.data.AUTOTUNE)
```

The `train` object is now of the following type

```
<ParallelMapDataset element_spec=(TensorSpec(shape=(256, 256, 3), dtype=tf.float32,
name=None), TensorSpec(shape=(3,), dtype=tf.float32, name=None))>
```

I wanted to play around with layers since I am new to neural networks and my dataset is not the same as in the example, so my model is not exactly like the one given in the link.

```
def make_model(input_shape, num_classes):
    inputs = keras.Input(shape=input_shape, batch_size=None)
    x = layers.Rescaling(1.0 / 255)(inputs)
    for size in [32,64,128]:
        x = layers.Conv2D(size, (2,2), strides=2, padding="same", input_shape =
input_shape)(x)
        x = layers.Activation("relu")(x)
        x = layers.MaxPooling2D(pool_size=(2, 2),padding='same')(x)

    activation = "softmax"
    units = num_classes
    outputs = layers.Dense(units, activation=activation)(x)
```

```
return keras.Model(inputs, outputs)
```

```
model = make_model(input_shape=(256,256,3), num_classes=3)
tf.keras.utils.plot_model(model, show_shapes=True)
```

Finally I compile and train the model

```
epochs = 25

callbacks = [
    keras.callbacks.ModelCheckpoint("save_at_{epoch}.keras"),
]
model.compile(
    optimizer=keras.optimizers.Adam(1e-3),
    loss="binary_crossentropy",
    metrics=["accuracy"],
)

model.fit(
    train,
    epochs=epochs,
    callbacks=callbacks
)
```

However, at this point I get the error from the title

```
ValueError: Input 0 of layer "model_12" is incompatible with the layer: expected shape=(None,
256, 256, 3), found shape=(256, 256, 3)
```

I don't know why the layers expect the extra `None` dimension and I don't know how to reshape the train object. I have tried reshaping using the answers given [here](#) but the type of my train object is not subscriptable.

Can someone explain me why and how I should make shapes match?

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asked Jan 1 at 12:54



Javi

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1 Model needs a batch dimension, therefore you need to set your batch size. That's how Keras works. – [Frightera](#) Jan 2 at 1:36

@Frightera thank you! I also needed to add a global average pooling layer to so that the output would match the dimension of the labels. – [Javi](#) Jan 2 at 12:02